LAB 16 - PROMPTS

Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab16-paper-trading-bot/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Build the risk layer and the kill switch (A7)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/risk.py with class RiskManager(cfg). kill_switch_on() = the file named by risk.kill_switch_file exists under labs. check(broker, status, symbol, wants_buy, allow_queue=False) returns (ok_to_trade, ok_to_buy, notes): kill switch -> (False, False, ['KILL_SWITCH present - trading halted']); session 'closed' -> blocked with 'market closed' unless allow_queue (then note 'order will queue (GTC) for the next session'); equity / last_equity - 1 <= -risk.max_daily_loss_pct -> no new buys; symbol not in risk.allowed_symbols -> no buy; account trading_blocked -> no buy; otherwise ['all checks passed']. cap_weight(weight, equity, price) applies max_position_pct and max_order_value. kill_main(argv) for `python -m pipeline kill --on | --off | --status`: --on writes the file with a timestamp; --off MOVES it to _state/kill_switch_history/KILL_SWITCH.<timestamp> (an audit record - never delete it). Risk is deterministic code; no model may override it. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline kill --status prints 'KILL SWITCH OFF'.

Step 2: Build one decision cycle (A7)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/runner.py. decide(bars, cfg) = the configured strategy's target from pipeline.signals.target_position on COMPLETED bars only, returning bar_date, close, signal and (for ema_cross) ema_fast and ema_slow. run_cycle(broker, om, risk, bars, run_time, cfg, weight, allow_queue=False, poll_seconds=0): read position and pending shares from the pipeline.execution.OrderManager; wants_buy = signal 1 and nothing held or pending; wants_sell = signal 0 and shares held; run RiskManager.check; if allowed call om.rebalance({symbol: weight}) or om.rebalance({symbol: 0.0}) with tag = the bar date (one decision per bar, so a double-fired job cannot double-trade); otherwise action BLOCKED if a trade was wanted; return a log row with run_time, bar_date, symbol, close, ema_fast, ema_slow, signal, position_before, action, qty, order_id, status, fill_price, equity, checks. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: run_cycle returns a dict with the 15 log fields.

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Step 3: Build the forward-test replay and the logs (A8)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/runner.py add replay(cfg, start, end): offline forward test on PaperBrokerSim (state_file bot_sim.json, reset) - for each 2025 session set the simulator clock to 09:35, mark the price at today's open, run_cycle on bars up to YESTERDAY's close with weight risk.max_position_pct, then process the day's bar and roll end_of_day; stop if the kill switch appears. Write _outputs/trade_log.csv (fresh, one row per session) and _outputs/Lab16_equity_log.csv (date, equity, position, close), print every non-HOLD row, the number of orders, start and end equity and SPY buy and hold over the same days, and save chart Lab16_forward_test.png. once(cfg): one cycle; offline it acts at the open after the last bundled bar and appends to trade_log.csv; online it appends to _outputs/live_paper_log.csv. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline run --help lists --once, --loop, --live-paper, --interval, --max-cycles, --start and --end.

Step 4: Add the live paper loop and scheduling (A7)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/runner.py add live_loop(cfg, interval, max_cycles): on the Alpaca PAPER account (or the simulator with a clear message if no keys) run up to max_cycles cycles: print the clock line, stop with 'KILL SWITCH ON - no orders sent; loop stopped' if the kill switch exists, run_cycle with weight live_paper.max_position_pct, allow_queue=True and poll_seconds live_paper.poll_seconds online (the OrderManager sends extended-hours limit orders outside regular hours), append to _outputs/live_paper_log.csv, print the row and every position with unrealised P&L, then sleep interval seconds; finish with the dashboard address. main(argv): --loop --live-paper -> live_loop; --loop offline -> replay; --loop online -> live_loop; otherwise once. schedule_main(argv) for `python -m pipeline schedule`: print and write to _outputs/Lab16_schedule/ a crontab line with CRON_TZ=America/New_York for schedule.run_at_exchange_time Mon-Fri (with the Singapore time in US summer and winter), a launchd plist (com.lioncitycapital.pipeline, WorkingDirectory labs) and a Windows schtasks command, each running `python -m pipeline run --once` from labs and appending to _outputs/scheduler.log. Do not install anything. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline schedule --help works.

Step 5: Run the 2025 forward test (A8, K7)

Do this (no prompt needed)
Run: python -m pipeline run --loop (same as: python3 Lab16-paper-trading-bot/run_stage.py --loop). Offline this replays 2025 day by day; each order fills at that day's open in the simulator.

Check: 250 sessions, 5 orders: BUY 159 after the 2025-01-21 close, SELL after 2025-02-27, BUY 166 after 2025-05-02, SELL after 2025-11-19, BUY 163 after 2025-11-28; equity USD 100,000.00 -> 116,249.79 (+16.25%) versus SPY buy and hold +17.06%; trade_log.csv with 250 rows; Lab16_forward_test.png.

Step 6: Audit the decision log (A8)

Prompt - copy into your AI coding assistant
Read labs_fullcode/_outputs/trade_log.csv and write an audit summary for the compliance officer: number of sessions, decisions by action, every order with its bar date, fill price and equity after the fill, and any row where a risk check did not say 'all checks passed'. Do not change any file.

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Check: 250 rows: 5 BUY/SELL rows with status filled and 245 HOLD rows; 249 rows say 'all checks passed' and one (2025-01-27) says 'daily loss -2.05% breaches 2% limit - no new buys' - a HOLD day with the position already held, so the rule blocked nothing.

Step 7: Test the kill switch (A7)

Do this (no prompt needed)
Run: python -m pipeline kill --on, then python -m pipeline run --once, then python -m pipeline kill --off.

Check: The --once row shows checks 'KILL_SWITCH present - trading halted' (no order can be sent); --off prints 'moved to _state/kill_switch_history/KILL_SWITCH.<timestamp>' - the halt is kept as an audit record.

Step 8: Run the strategy LIVE on the paper account (A7, A8, K7)

Do this (no prompt needed)
With paper keys in .env, during class run: python -m pipeline run --loop --interval 60 --live-paper --max-cycles 5. Watch decision -> order -> fill -> position -> log for each cycle. After two or three cycles, in a SECOND terminal run python -m pipeline kill --on and watch the next cycle halt. Then run python -m pipeline kill --off and open _outputs/live_paper_log.csv and the Alpaca dashboard.

Check: Cycle 1 sends one SPY buy for 10% of equity (an extended-hours LIMIT DAY order in Singapore class hours; about 14 shares on USD 100,000) and prints its lifecycle and the position; later cycles HOLD ('buy order ... still working' or position held); after kill --on: 'KILL SWITCH ON - no orders sent; loop stopped'. Without keys the same loop runs on the simulator (at a weekend the buy queues as a GTC limit order).

Step 9: Schedule the runner (A7)

Do this (no prompt needed)
Run: python -m pipeline schedule and read the three entries. Only install one if your instructor approves (crontab -e, launchctl load, or schtasks in a Command Prompt).

Check: One cycle at 09:35 New York = 21:35 Singapore (US summer) / 22:35 (US winter), Mon-Fri; files crontab.txt, com.lioncitycapital.pipeline.plist and schtasks.cmd in _outputs/Lab16_schedule/; nothing is installed.

Run it

Step	Command / file
Run Lab 16	python3 Lab16-paper-trading-bot/run_stage.py (same as: python -m pipeline run)
The complete code	pipeline/risk.py, pipeline/runner.py (reference copy: Lab16-paper-trading-bot/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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